

PROJECT DOCUMENTATION PHASE

Final Report

Project Title

Optimization of Users, Groups, Roles with Access Control and Workflows in ServiceNow

Abstract

The project "Optimization of Users, Groups, Roles with Access Control and Workflows" was developed using the ServiceNow platform to improve identity and access management within an organization. The solution automates user management, group organization, role assignment, Access Control Lists (ACLs), and approval workflows to provide a secure and efficient system.

Project Summary

The project focuses on simplifying administrative tasks by centralizing user and access management. Manual processes were replaced with automated workflows to reduce errors, improve security, and increase productivity. ServiceNow features such as User Administration, Group Management, Role-Based Access Control (RBAC), ACLs, and Flow Designer were used to implement the solution.

Objectives Achieved

- Successfully created and managed users.
- Created and maintained user groups.
- Assigned roles based on responsibilities.
- Configured Access Control Lists (ACLs).
- Automated approval workflows using Flow Designer.
- Improved security through role-based access control.
- Reduced manual administrative effort.

Modules Implemented

- User Management
- Group Management
- Role Management
- Access Control Lists (ACLs)

- Workflow Automation
- Notification Management

Testing Summary

The application was tested for:

- User creation
- Group creation
- Role assignment
- ACL validation
- Workflow execution
- User Acceptance Testing

All modules functioned successfully and met the expected requirements.

Results

The implemented solution:

- Improved access security.
- Reduced manual work.
- Increased efficiency.
- Automated approval processes.
- Enhanced audit and compliance capabilities.

Future Enhancements

- Multi-level approval workflows.
- Integration with external identity providers.
- Advanced reporting dashboards.
- Mobile approval support.
- AI-based access recommendations.

Conclusion

The project successfully demonstrates how ServiceNow can optimize user, group, and role management using Access Control Lists and automated workflows. The developed solution provides secure, scalable, and efficient access management suitable for modern organizations.

Functional Specification Document (FSD)

1. Purpose

The purpose of this Functional Specification Document is to describe the functional requirements, system behavior, and implementation details of the ServiceNow application developed for optimizing user, group, role, access control, and workflow management.

2. Project Scope

The application provides:

- User creation and management.
- Group management.
- Role assignment.
- Access Control List (ACL) configuration.
- Workflow automation using Flow Designer.
- Approval and notification management.

3. Functional Requirements

User Management

- Create new users.
- Update user information.
- Manage user profiles.

Group Management

- Create groups.
- Assign users to groups.
- Modify group membership.

Role Management

- Create roles.
- Assign roles to users.
- Remove or update roles.

Access Control

- Configure ACL rules.
- Restrict unauthorized access.
- Validate user permissions.

Workflow Management

- Trigger approval workflows.
- Send notifications.
- Track request status.

4. Input

- User information
- Group details
- Role details
- Access request
- Approval information

5. Output

- User account creation
- Role assignment confirmation
- Workflow approval status
- Notification messages
- Secure user access

6. Security Features

- Role-Based Access Control (RBAC)
- Access Control Lists (ACLs)
- User authentication
- Authorization checks
- Audit logging

7. Assumptions

- Users have valid login credentials.
- Administrators manage role assignments.
- Workflow approvals follow organizational policies.

8. Constraints

- Only authorized administrators can modify roles.
- ACLs determine record-level access.
- Workflow execution depends on approval conditions.

9. Conclusion

The Functional Specification Document defines the complete functional behavior of the ServiceNow application and ensures that all project requirements are implemented accurately and securely.